Turret Test

Calibrate the turret IN ORDER: servo direction, encoder direction, gear ratio, limits, aiming. Then the hood and the home switch. (issue #31, decisions 019, 021, 033)

Names		Date	
Battery voltage at start		Battery at end	

OpMode: Turret Servo Raw Test, then Turret Test (Tests group on the Driver Station). **Needs in the config:** `turretRotateServo` (Continuous Rotation Servo), `turretEncoder` (Motor port, encoder only), `turretHomeSwitch` (Digital Device), `turretHoodServo` (Servo)

Hidden by default (052): delete `@Disabled` in its file, build, deploy. **Safety:** space around the robot, hands clear, one person on STOP. Nothing moves in INIT.

Nothing physical stops the turret. Until step 4 is done it can turn into wires or the frame. Keep a hand near STOP. Turret angles: 0 = the MAGNET = robot forward, + = LEFT. Limits: 120.0 / -120.0 (decisions 047, 048).

Part A: Turret Servo Raw Test (servo alone)

#	Do this	Expected	Actual	If wrong
A1	Push the left stick left and hold	turret KEEPS turning while held, stops when let go		moves a bit, stops, comes back = servo still in positional mode: reprogram it as continuous
A2	Stick centered, "Speed 0.00"	turret does not creep		recenter with the servo programmer

Controls (gamepad 1)

Button	Does
HOLD dpad LEFT / RIGHT	turn left / right (half speed). + right trigger = full speed (once homed). + bumper = slow
Y	set a MARK: "Since mark" counts degrees from here
A	turn to the REST spot (the home magnet = forward). Not homed yet? It turns LEFT to find the home switch first. Do steps 1 to 3 first!
X / B	turn to 45° left / right of ROBOT FORWARD (same warning)
Dpad UP / DOWN	hood up / down a little
BOTH lights GREEN	turret is at HOME: the switch (forward), or within HOME_TOLERANCE_COUNTS of it

Part B: Turret Test, steps IN ORDER

#	Do this	Expected	Actual (what happened)	If wrong
1	Hold dpad LEFT	turret turns LEFT (counter-clockwise from above)		`SERVO_REVERSED = true`

#	Do this	Expected	Actual (what happened)	If wrong
2	Turn left, watch "Turret angle"	angle goes UP		`ENCODER_REVERSED = true`
3	Note "Turret angle", then turn exactly 90° left (use a square or tape marks)	it changes by about 90		new `ENCODER_REVS_PER_TURRET_REV` = old value x (change it read) / 90 = _____
4	Home it. Turn slowly left until just before anything hits. Read the angle. Same to the right	left: ____° right: ____° (turret angles)		set `MAX_LEFT_DEGREES` / `MAX_RIGHT_DEGREES` a few degrees inside those
5	Turn past the home switch (forward) once (homes). Press A, then X, then B	turns to the magnet, then 45 L / 45 R of forward, no wobble		wobbles: lower `AIM_GAIN`. Too slow: raise it. Runs away: steps 1 or 2 are wrong
6	Turn through the home switch slowly, watch "Switch raw reading"	reading flips at the magnet		flips the other way: `HOME_SWITCH_TRIGGERED_STATE` to the other value
7	Dpad DOWN until the hood is flattest, read the position. Then UP to steepest	lowest: ____, highest: ____		set `HOOD_LOWEST_POSITION` / `HOOD_HIGHEST_POSITION` just inside the hard stops

Numbers to put into the code: `subsystems/Turret.java`

Constant	Was	New value	Checked by
`SERVO_REVERSED`	false		
`ENCODER_REVERSED`	false		
`ENCODER_REVS_PER_TURRET_REV`	4.11		
`FORWARD_ANGLE_DEGREES` (magnet = 0)	0.0		
`MAX_LEFT_DEGREES` / `MAX_RIGHT_DEGREES`	120.0 / -120.0		
`AIM_GAIN`	0.02		
`HOME_SWITCH_TRIGGERED_STATE`	false		
`HOOD_LOWEST_POSITION` / `HOOD_HIGHEST_POSITION`	0.0 / 1.0		
`HOME_TOLERANCE_COUNTS`	100		

After the test: a coder puts the new numbers into the constants and removes the matching TODO comments; the results go on GitHub issue #31. Photos of anything surprising!